CV

AI CAD-to-Cost Analysis

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AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

87

Machined Block with Two Through-Holes

60 × 40 × 20 mm · 44.9 cm³ · Al 0.121 kg / Steel 0.352 kg

Manufacturability: 87/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	**60.0 × 40.0 × 20.0 mm**	Surface Area	**97.4 cm²**
Volume	**44.86 cm³**	Weight (Al)	**0.121 kg**
Weight (Steel)	**0.352 kg**	Weight (Plastic)	**0.047 kg**

§2 — Detected Features

Feature Type	Count	Significance	Description
Through-hole (cylindrical)	2	High	2× drilled holes, r=5mm, likely dowel/clearance holes
Planar prismatic faces	6	Medium	6 orthogonal planar faces forming a rectangular block
Zero-draft walls	6	Low	All 6 primary faces at 90° (no draft) — consistent with machined-from-solid part, not cast/moulded

§3 — Material Analysis

AI-suggested

Primary Material: **Mild/Low-Carbon Steel (DC01-equivalent bar stock) (55% confidence)**

No aluminium signal present (no lightweighting note, no thin HPDC-style wall, no Al hint in filename). Per guidance, a dense, thick-section solid block (min wall 9.95mm, near-solid fill ratio 0.93) with no explicit lightweight material call-out defaults to steel over aluminium. Chosen mild steel (DC01-equivalent, density 7.85 g/cm³) as the generic machinable steel rather than stainless, since no corrosion/food-grade signal exists either.

Alternatives: **Aluminium 6061-T6 (30%)** · **Stainless Steel 304 (15%)**

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
CNC Milling (3-axis, 3 setups)	machining	90%	0.7950	User-forced machining process. Geometry is simple prismatic block with 2 drilled holes across 3 approach directions (+X:4f, +Y:2f, +Z:2f), consistent with a 3-axis VMC job requiring 3 setups.

§5 — Manufacturability Risks (Score: 87/100)

Severity	Feature / Area	Description	Recommended Action
Medium	**3 orthogonal setups (+X,+Y,+Z)**	Each re-orientation introduces alignment/stack-up error and adds non-cutting setup time (0.75hr of 0.795hr total, ~94%)	Consolidate features to 2 setups using a sub-plate/vice with indexing, or use a 4/5-axis machine to cut setup count.
Medium	**Wall thickness min=9.95mm near hole**	Minimum section thickness is close to hole diameter (10mm), risking distortion/chatter during drilling near thin section	Verify hole edge distance from wall boundary and consider peck drilling or reduced feed near thin sections.
Low	**2× Ø10mm holes**	Standard drill-size holes, low risk but require tight positional tolerance if used for dowel/location	Specify reamed/bored tolerance only if functionally required to avoid unnecessary secondary operations.

§6 — DFM Issues (machining)

Severity	Area	Description	Impact	Fix
Medium	**Setup count (3 orientations)**	Part requires re-orientation across +X, +Y, +Z faces (4+2+2 features), driving setup time to 94% of total cycle time	At 200,000 units/year, setup dominates non-value-add time unless heavily amortized via fixturing/pallet systems	Design features to be accessible from 2 or fewer faces, or invest in multi-axis fixturing/pallet automation for high volume
Medium	**Thin section near hole (9.95mm min wall)**	Minimum wall thickness measured near a hole is close to the hole diameter, increasing risk of deflection/chatter	Potential dimensional inaccuracy or surface finish issues on the thin wall during drilling/milling	Increase wall-to-hole edge distance or reduce feed/speed locally; consider support fixturing at that wall
Low	**Hole tolerance/position**	If holes function as locating/dowel features, 3-setup machining increases risk of positional stack-up error between orientations	Could require secondary boring/reaming operation to hold tight true-position tolerance	Machine both holes in a single setup if possible, or specify realistic (non-precision) tolerance if not functionally critical
Low	**Stock utilization**	Near-solid block (fill ratio 0.93) machined from bar stock removes only ~7% of material, mostly to hole bores	High material utilization is good for cost, but bar-stock sizing should match bounding box tightly to avoid excess stock cost	Source near-net bar stock (60x40mm cross-section) to minimize scrap

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£2.60	**£3.25**	**£4.80**

Net Weight	**0.352 kg**	Material	**steel**
Recommended Process	**machining**	Cycle Time	**0.0450 hr/part**

Setup Time	**1.000 hr**	Operations	**4 ops**
Operations Detail	**Milling — +X (4 faces) (mach-haas-vf2, 0.0133 hr) Milling — +Y (2 faces) (mach-haas-vf2, 0.0067 hr) Milling — +Z (2 faces) (mach-haas-vf2, 0.0067 hr) Drilling — 2 holes (2×Ø10.0×20) [geometry-measured] (mach-drill, 0.0183 hr)**		

Process-Specific Parameters

Casting Subtype	**sand**
Die/Mould Cost	**£2,500**
Die Life	**500 shots**
Cavities	**1**
Yield	**70.0%**
Flash Weight	**0.050 kg**
Yield	**85.0%**
Die Cost	**£15,000**
Strokes	**6**
Cavities	**0**
Wall Thickness	**0 mm**
Mould Cost	**£0**
Mould Life	**0 shots**
Projected Area	**0.0 cm²**

§8 — AI Analysis Explanation

This is a simple prismatic steel block (60x40x20mm) with two Ø10mm through-holes, machined from bar stock in 3 setups (+X, +Y, +Z) per the deterministic geometry analysis. Material was inferred as mild steel (DC01-equivalent) rather than aluminium because no lightweighting or thin-wall HPDC signal is present, and the guidance directs toward steel for dense, thick-section, unspecified metal parts. Cycle time (0.795hr total, of which 0.75hr is setup across 3 orientations) and manufacturability score (87/100) are used verbatim from the OCCT-derived bottom-up estimate. At 200,000 units/year, setup time should be amortized across batches in the actual cost model; unit cost estimate reflects this with setup partially amortized rather than charged in full per part.

§9 — Analysis Limitations & Assumptions

1. Material selection (steel vs aluminium) is inferred from guidance heuristics, not from explicit metadata, drawing notes, or filename — flagged as UNDECIDED in the brief.
2. Hole depth (through vs blind) and functional tolerance requirements not specified in geometry data, affecting DFM risk assessment.
3. Cost range assumes amortized setup time at typical high-volume batch sizes (200–500 units per changeover); actual batching strategy may shift the unit cost.
4. No thread/tap features detected; if either hole is intended to be threaded, cycle time and tooling would need revision.

Stage-1 pre-selection: auto (0%) ·